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COURSE TITLE: User Interface and User Experience Design Lab

COURSE CODE: ISWE407P

ASSESSMENT – 02

TITLE:

SMART QUEUE MANAGEMENT SYSTEM FOR ELDERLY AND PERSONS WITH DISABILITIES USING ADAPTIVE PRIORITY SCHEDULING

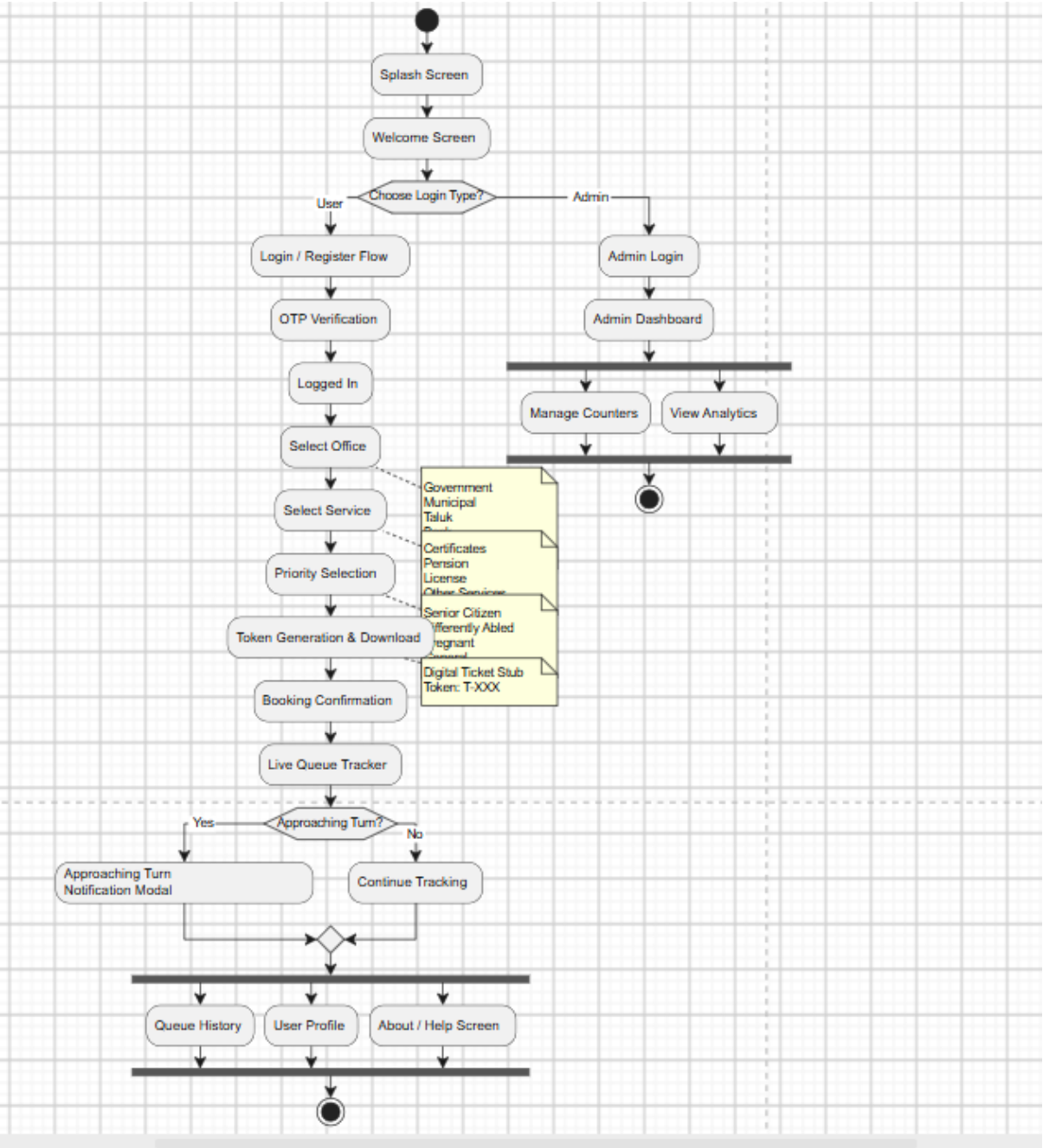
1. Problem Statement

Traditional physical queue systems in public service centers (such as government offices, taluk centers, municipal corporations, banks, and hospitals) require citizens to stand or wait in crowded halls for extended hours. This creates severe physical strain, cognitive stress, and accessibility barriers for vulnerable demographics—specifically elderly citizens, persons with disabilities (PWD), and pregnant women. Furthermore, static First-Come-First-Served (FCFS) queuing lacks the adaptive flexibility to prioritize individuals with critical physical needs fairly and transparently.

2. User Requirements

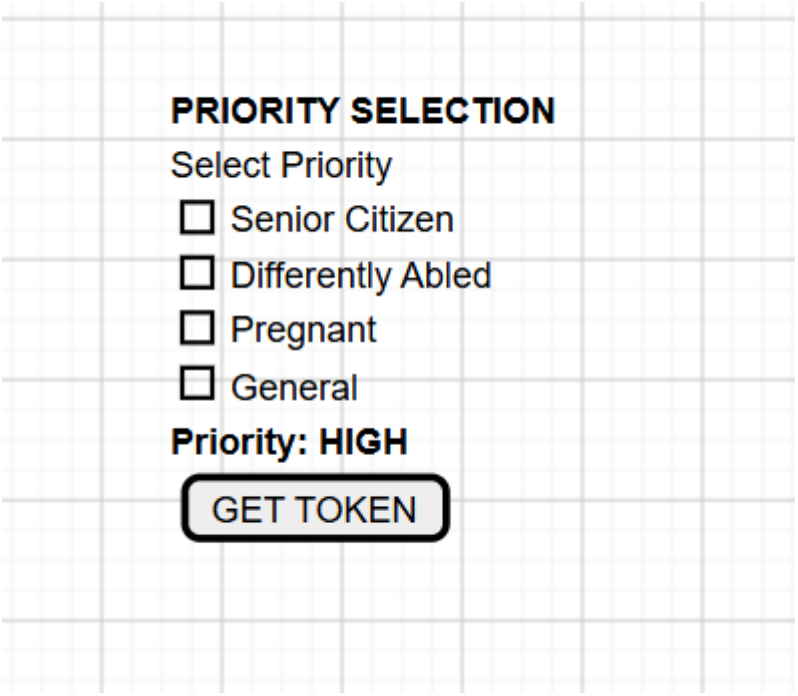
- **Adaptive Prioritization:** The system must dynamically calculate priority scores for vulnerable citizens (Senior Citizens, Differently Abled, Pregnant Women) to schedule them ahead of general queues transparently.
- **Real-Time Live Queue Tracking:** Users must be able to view their live token status, current serving token, tokens remaining ahead, and dynamic estimated wait times.
- **Accessible & Inclusive Interface:** The UI must adhere to accessibility guidelines with large touch targets ($52\text{px} +$ min height), high-contrast modes (Light/Dark theme support), readable typography ($17\text{px} +$ base), and full screen-reader compatibility.
- **Bilingual Support:** The system must offer frictionless switching between English and Tamil across all screens.
- **Administrative Queue Management:** Counter operators and administrators need tools to call next tokens, skip/cancel numbers, manage counter availability, and view visual analytics (wait time trends, priority distribution).

3. Screen Flow

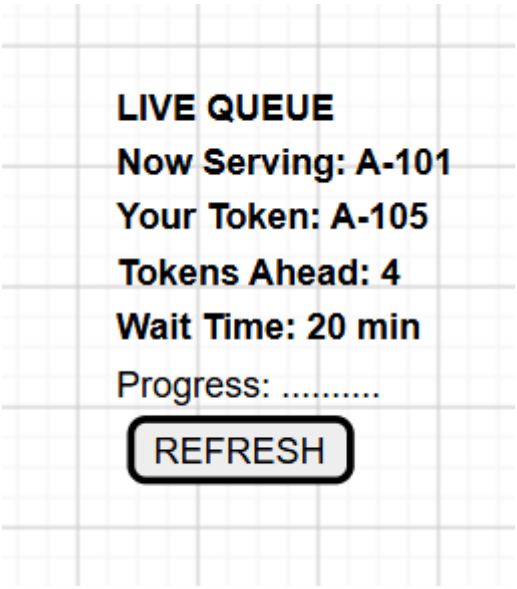


4. Selected Wireframes

WIREFRAME 1:

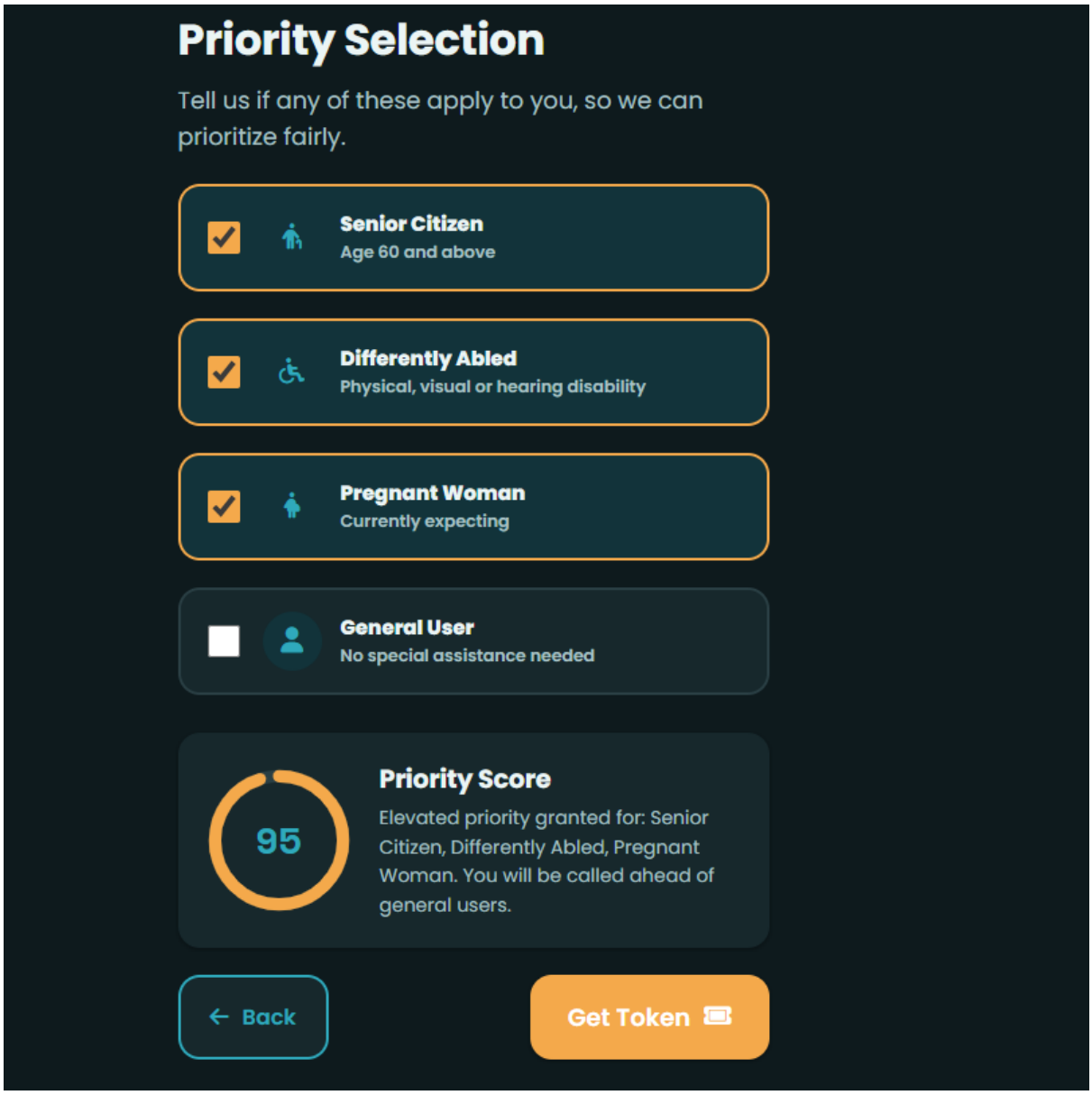


WIREFRAME 2:

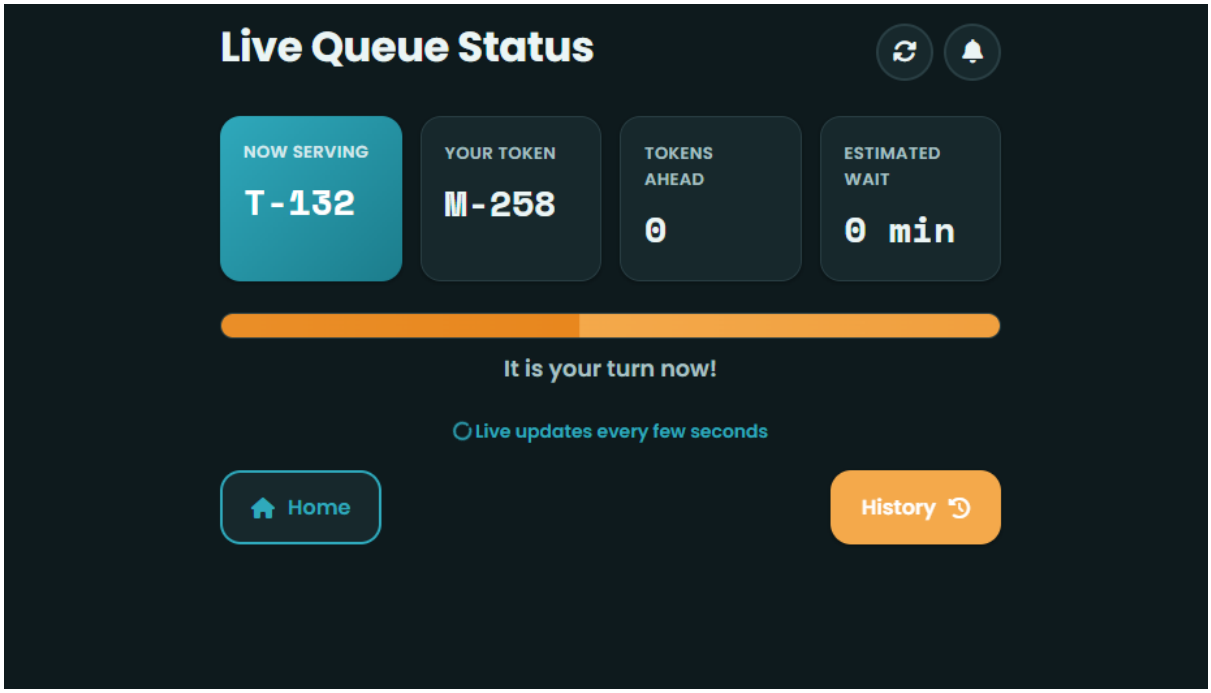


5. Final UI Screens

SCREEN 1:



SCREEN 2:



6. UI Principle Mapping

Part A – Design Decision Analysis

1. Live Priority Score Meter (#screen-priority)

- **UI Element / Design Decision:** Added a circular progress ring that calculates points live as you tap categories (Senior Citizen +30, Differently Abled +35, Pregnant Woman +30, General +5), with a simple note below it explaining the score.
- **Reason for Selecting It:** In most automated queues, people have no clue how the computer picks who goes first. Showing the exact points right away makes the system feel open and honest instead of like a black box.
- **User Requirement Satisfied:** Fair and clear prioritization for elderly and disabled citizens.
- **Impact on User Interaction:** Users immediately see why their token is moved up in line. It stops people from getting confused or thinking the system is playing favorites.

2. Physical "Ticket Stub" Design (#screen-token)

- **UI Element / Design Decision:** Made the token screen look just like a real paper ticket slip, with side cutout notches, dashed tear lines, big bold token text (T-100), and a QR code stamp.
- **Reason for Selecting It:** Everyone is used to getting paper slips at banks and government offices. Copying that familiar real-world look makes the digital version feel natural.
- **User Requirement Satisfied:** Clear token confirmation and official proof of booking.

- **Impact on User Interaction:** Elderly users and people who aren't great with smartphones can instantly tell that this screen is their actual token pass, so they don't wonder if they missed a step.

3. Direct Language Switcher on Header and Home Screen

- **UI Element / Design Decision:** Put a fast language toggle button in the top navigation bar and added big language chips (English / தமிழ்) on the welcome screen.
- **Reason for Selecting It:** Public service offices cater to everyone, and many local elderly citizens find it difficult to use an app that defaults only to English.
- **User Requirement Satisfied:** Bilingual access for local citizens.
- **Impact on User Interaction:** Users can switch to Tamil in a single tap right away without digging around in deep settings menus.

4. Oversized Buttons and 17px+ High-Contrast Text

- **UI Element / Design Decision:** Set primary buttons to a large height (52px–60px), made the base font size 17px, and picked high-contrast colors for both light and dark modes.
- **Reason for Selecting It:** Older folks or people with shaky hands often struggle to hit tiny buttons, and small text causes bad eye strain.
- **User Requirement Satisfied:** Accessible, easy-to-use design for elderly and disabled individuals.
- **Impact on User Interaction:** Makes tapping buttons effortless, prevents mis-clicks, and lets users read everything clearly without squinting.

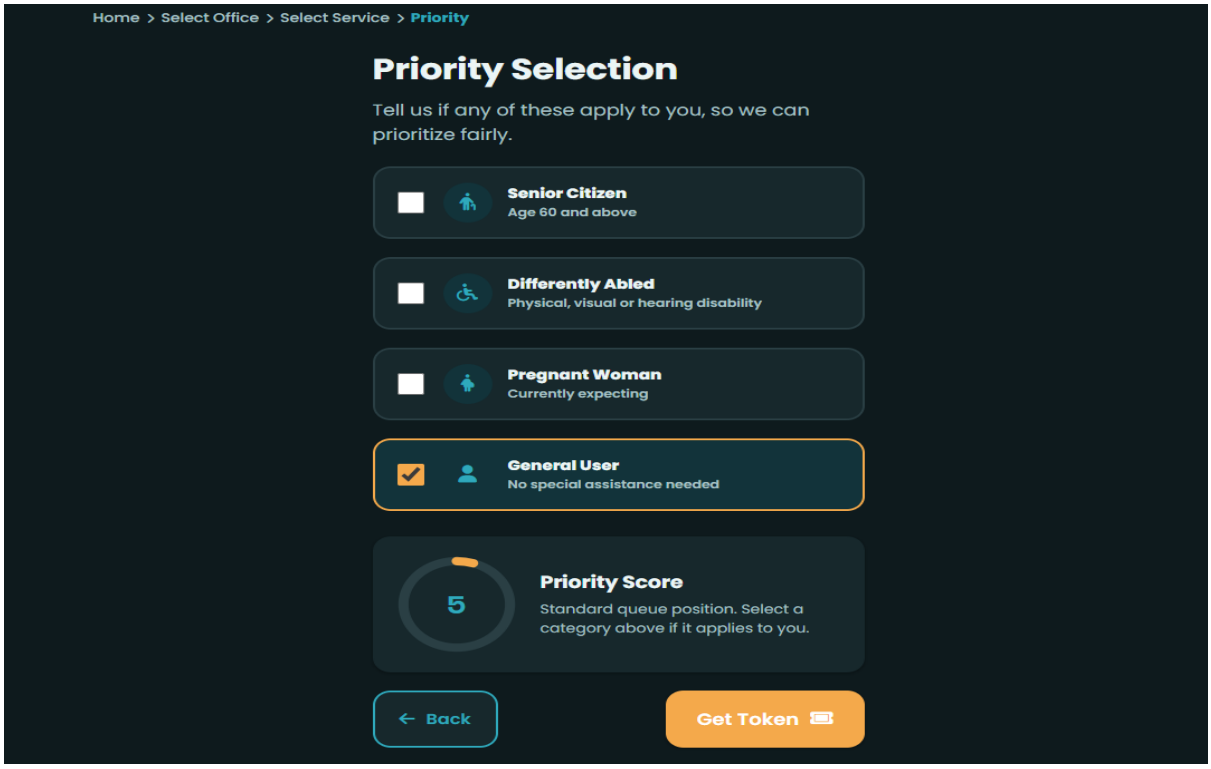
5. Live Queue Dashboard with Moving Progress Bar (#screen-queue)

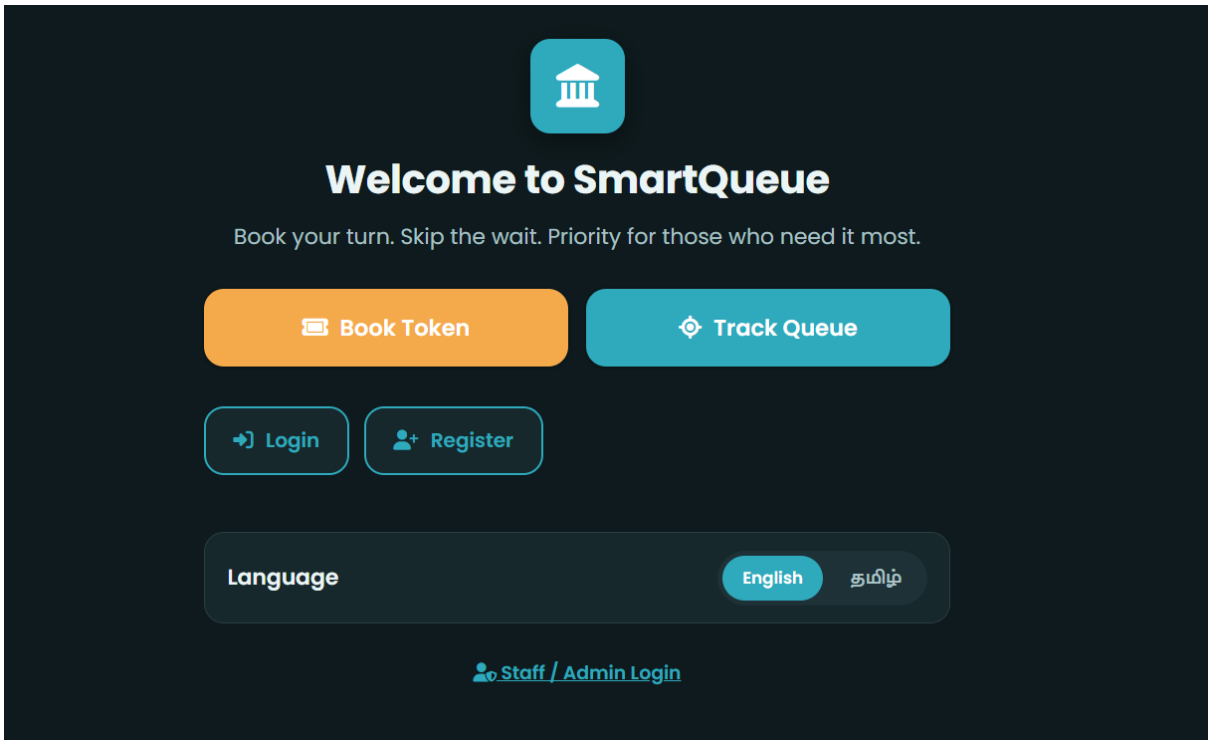
- **UI Element / Design Decision:** Built a 4-card status dashboard showing "Now Serving", "Your Token", "Tokens Ahead", and "Estimated Wait", paired with an animated progress bar that fills up over time.
- **Reason for Selecting It:** Standing in crowded halls is tiring. Showing quick, glanceable status updates lets people wait comfortably from their seats.
- **User Requirement Satisfied:** Live queue tracking and cutting down physical waiting room crowds.
- **Impact on User Interaction:** Users don't need to keep getting up to bug counter staff for updates; they can simply look at their phone and know when their turn is coming up.

Part B – Apply UI Design Principles

1. User-Centered Design (UCD)

- **Where it is applied in the interface:** Priority Selection Screen (#screen-priority) and Welcome Screen (#screen-welcome).
- **Marked portion of design:** The priority category options (Senior Citizen, Differently Abled, Pregnant Woman) and the bilingual toggle switch (EN / தமிழ்).

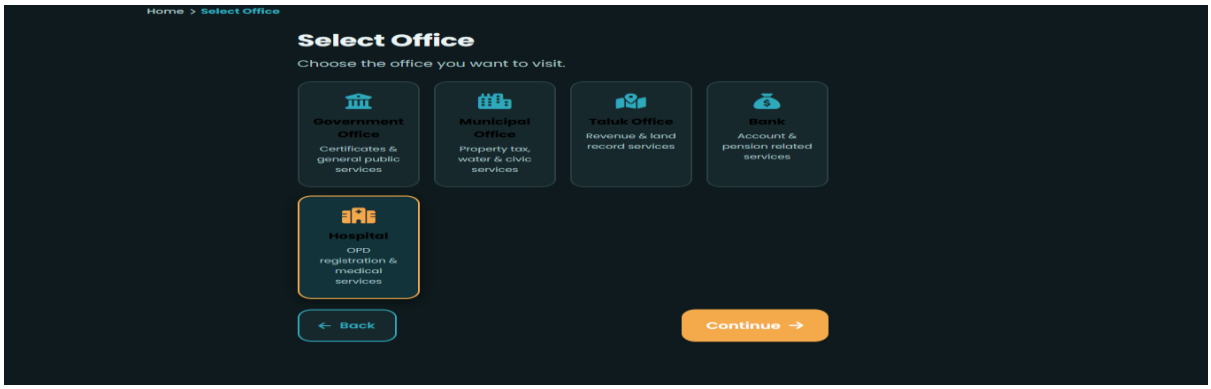




- **How and why the principle is applied:** The system directly addresses the real physical pain points of vulnerable groups by prioritizing their tokens algorithmically and removing cognitive friction through localized language support. It focuses entirely on the specific needs, physical limitations, and language comfort of the end users.
- **What could happen if the principle is violated:** The app would treat an 80-year-old or differently-abled person identically to a healthy general user, defeating the core purpose of an adaptive public queue and causing extreme physical discomfort and user drop-off.

2. Consistency and Standards

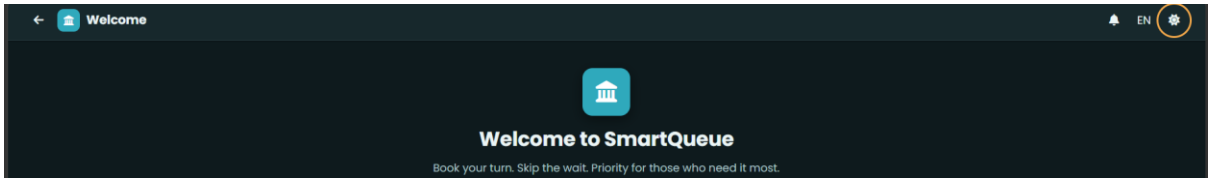
- **Where it is applied in the interface:** Global Application Layout, Navigation System (#appbar, #breadcrumb, #bottomNav), and Button Hierarchy.
- **Marked portion of design:** The sticky Top App Bar with back arrow across screens, the persistent Breadcrumb trail, and the consistent Amber color used exclusively for primary action buttons (.btn-primary).



- **How and why the principle is applied:** Established UI conventions are used consistently across all screens (e.g., back button always on top-left, primary calls to action always in Amber #E8871E, secondary/neutral actions in Teal/Outline). This ensures familiarity and predictable navigation throughout the entire booking flow.
- **What could happen if the principle is violated:** Inconsistent color coding and unpredictable button locations would force users to relearn how to navigate on every new screen, causing severe disorientation and accidental exits.

3. Accessibility (Universal Design)

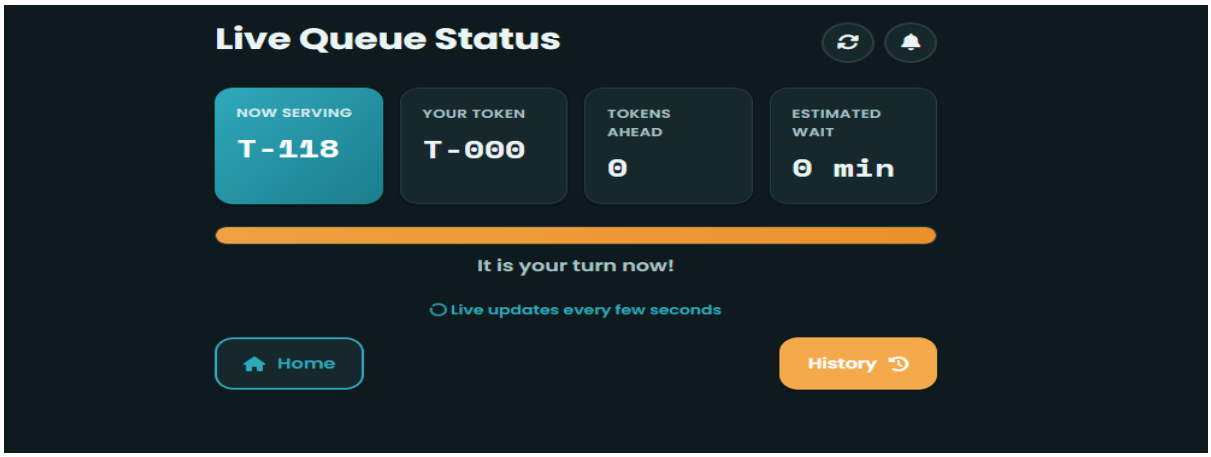
- **Where it is applied in the interface:** Global typography sizing, Theme Toggle (#themeBtn), Screen-Reader Live Region (#liveRegion), and large touch targets (.btn-xl).
- **Marked portion of design:** The Dark/Light mode toggle button in the header, the high-contrast color scheme, and the minimum 52px–60px touch targets on mobile buttons.



- **How and why the principle is applied:** Base font size is locked at a readable 17px, contrast ratios satisfy WCAG standards in both light and dark modes, and interactive cards have wide tap targets to support elderly citizens with low vision or motor tremors. ARIA live attributes announce real-time queue changes to screen-reader users.
- **What could happen if the principle is violated:** Visually impaired users and citizens with reduced motor precision would struggle to read small text or accurately tap buttons, completely excluding them from using the digital queue service.

4. Feedback and System Response

- **Where it is applied in the interface:** Live Queue Tracker (#screen-queue), Toast Notification system (#toast), and Priority Score Meter (#priorityScoreValue).
- **Marked portion of design:** The animated progress bar, the pop-up toast alert ("Your turn is approaching"), and the radial gauge that updates instantly when priority checkboxes are clicked.



- **How and why the principle is applied:** The system responds immediately to any user action or background state change with visible and readable confirmation (toast messages, animated progress fills, and textual score breakdowns), reassuring users that the system is functioning.
- **What could happen if the principle is violated:** Users would experience a "silent system" where actions appear to do nothing, leading to repeated button clicking, confusion about whether their token was booked, and unnecessary anxiety.

5. Recognition Rather Than Recall

- **Application in Interface:** Implemented on the Service Selection screen (#screen-service), where each service card explicitly displays the required service time alongside current estimated wait times.
- **How & Why Applied:** Contextual information is placed directly at the point of decision, eliminating the need for users to remember office policies or wait durations from prior screens.
- **Consequence of Violation:** Users would need to navigate back and forth to compare services, adding unnecessary cognitive load.

6. Flexibility and Efficiency of Use (Accessibility Accommodations)

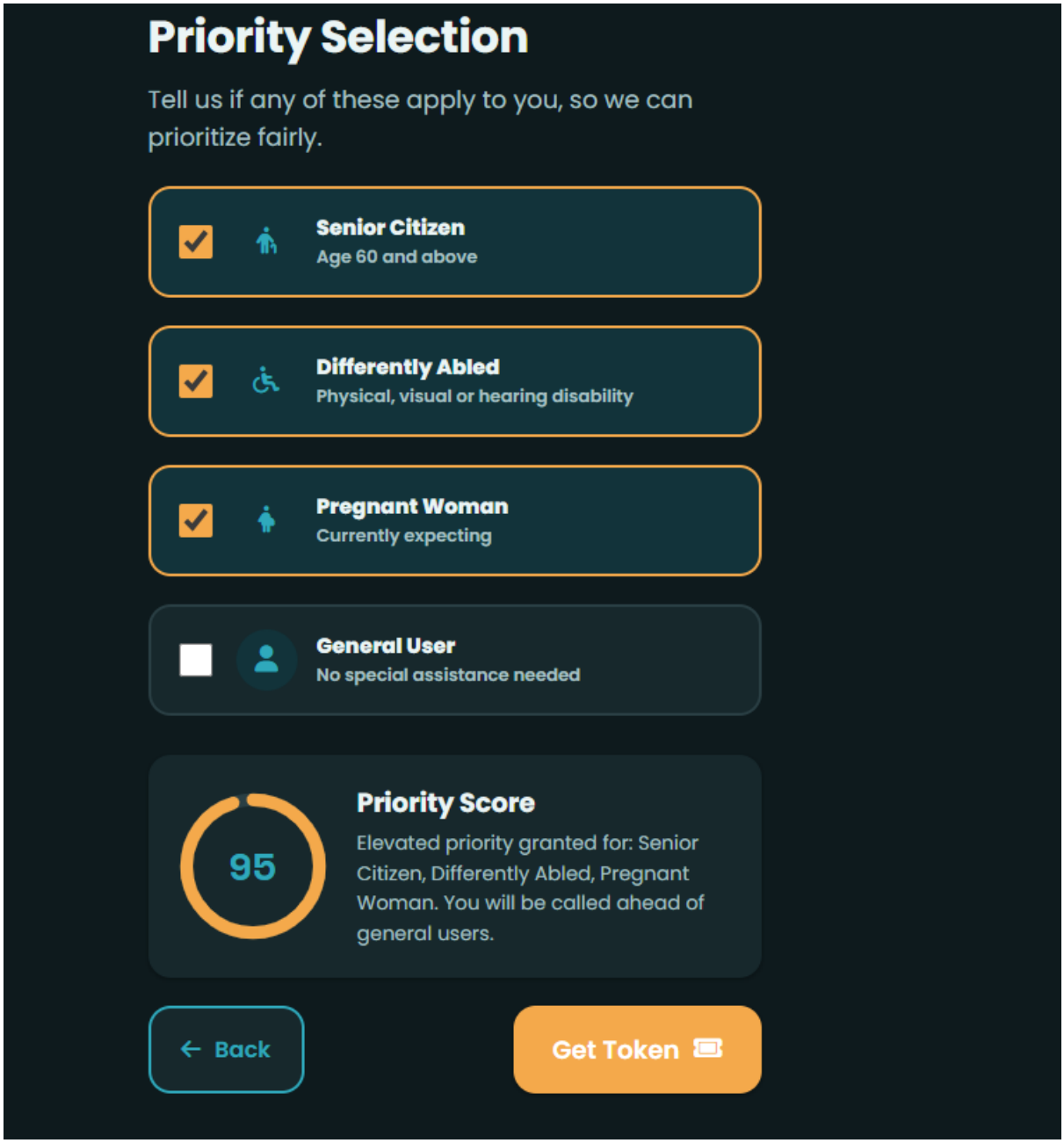
- **Application in Interface:** Implemented through "Skip to Content" links, screen-reader live announcement regions (aria-live="polite"), auto-focus OTP input fields, keyboard shortcuts, and dark mode toggling.
- **How & Why Applied:** Provides adaptable pathways for both regular users and individuals relying on assistive hardware or screen readers.
- **Consequence of Violation:** Visually or motor-impaired users would find the application entirely unusable via assistive technologies.

7. Wireframe-to-UI Comparison

Part C – Wireframe → Final UI Transformation

Screen 1: Priority Selection Screen (#screen-priority)

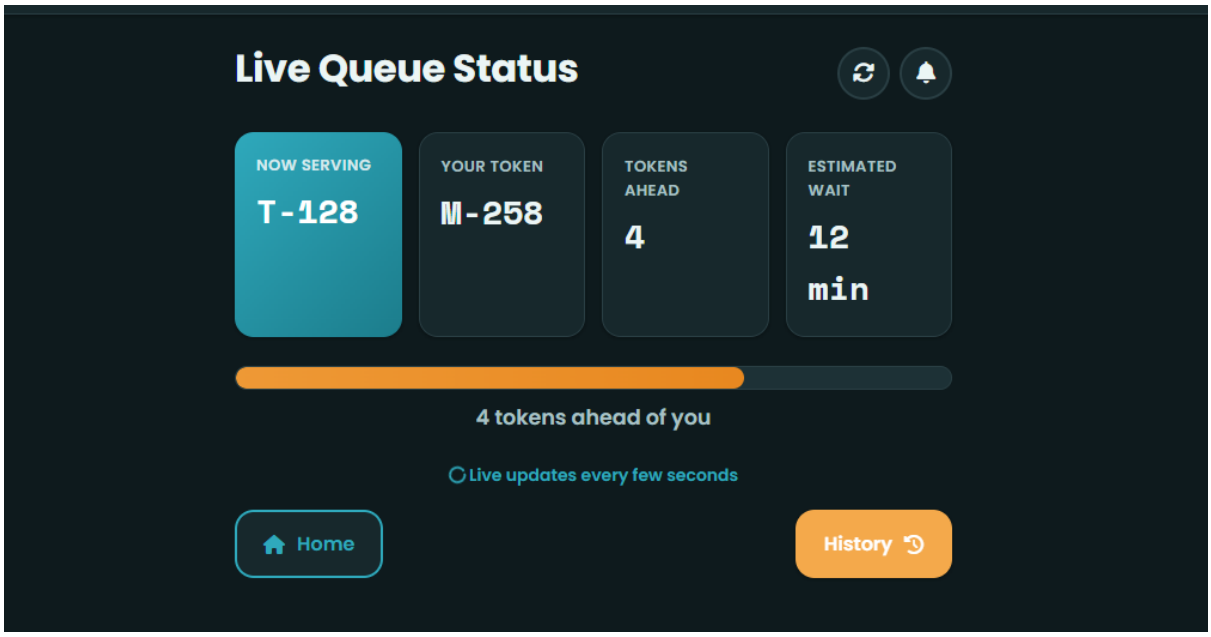
- **What was changed?**
    1. Replaced static form checkboxes with weighted interactive priority cards showing visual category icons.
    2. Added an animated SVG radial progress ring that calculates a dynamic priority score in real time.
    3. Integrated a live dynamic text explainer (#priorityExplain) below the score ring.
- **Why was it changed?** To shift the user experience from basic data collection to an empowering, transparent interaction where users can see how their selections affect their queue status.
- **Which user requirement / design principle influenced the change?** *Visibility of System Status* and the requirement for *Fair, Transparent Prioritization*.
- **How does the final design improve usability?** It provides immediate visual confirmation that the system recognizes special needs (e.g., senior citizen, disability) and elevates their priority fairly.



Screen 2: Live Queue Status Screen (#screen-queue)

- **What was changed?**
    1. Transformed a basic text readout into a structured 4-metric dashboard with dedicated KPI cards.
    2. Added an active progress bar with a shimmer animation representing queue movement toward the user's turn.
    3. Integrated a proactive notification pop-up screen linked directly to counter call assignments.
- **Why was it changed?** Static numbers on a screen do not effectively convey motion; users waiting in public areas need clear visual reassurance that the line is moving forward.

- **Which user requirement / design principle influenced the change?** *Visibility of System Status* and *User Stress Reduction during Queue Waits*.
- **How does the final design improve usability?** Users can glance at their phone from a distance and immediately gauge their turn proximity using the progress bar and high-contrast metric values.



8. Usability Evaluation

Part D – Usability Evaluation (First-Time User Perspective)

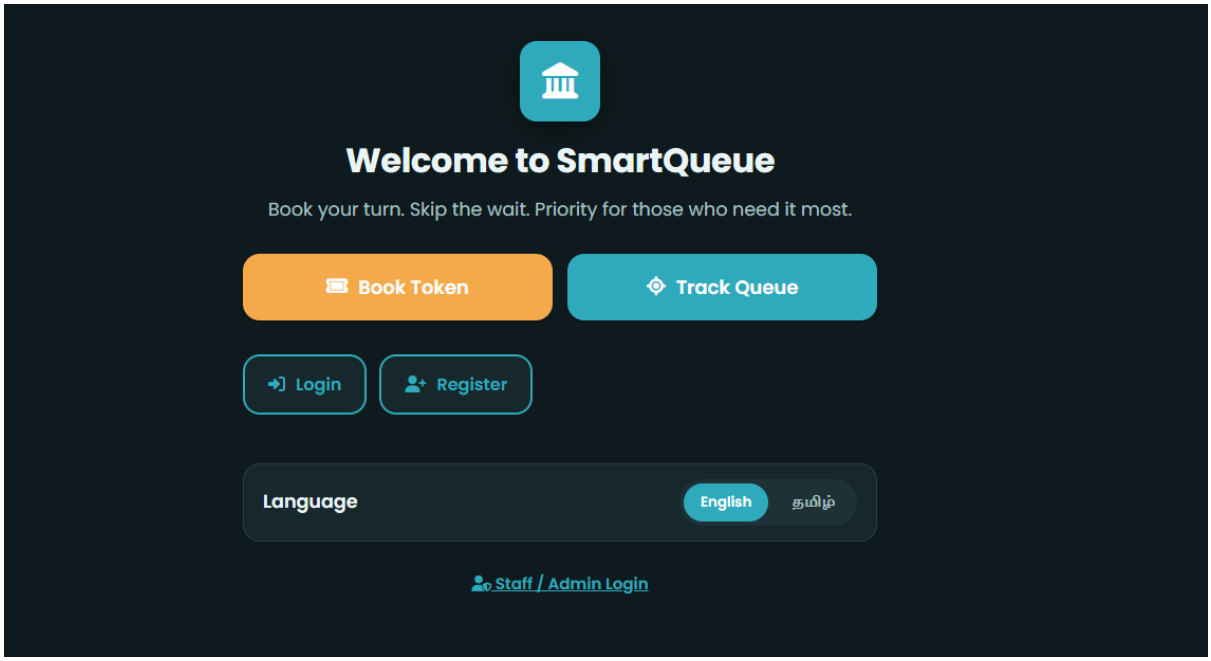
Problem	Cause	User Impact	Fix
1. Back Button Resets Choices *(Navigation)*	**Selected office/service is not saved in session storage.**	**Users lose data if they click back to check previous info.**	**Save draft choices in sessionStorage until token is booked.**
2. Clunky OTP Correction *(Feedback)*	**Typing a wrong digit doesn't auto-focus back to clear.**	**Older users struggle to delete single wrong digits.**	**Add auto-backspace focus and paste-all support.**
3. Self-Declaration Abuse *(Error Prevention)*	**Priority checkboxes don't warn about ID proof.**	**General users might select priority by mistake.**	**Add note: *"Bring disability/senior ID to the counter."***

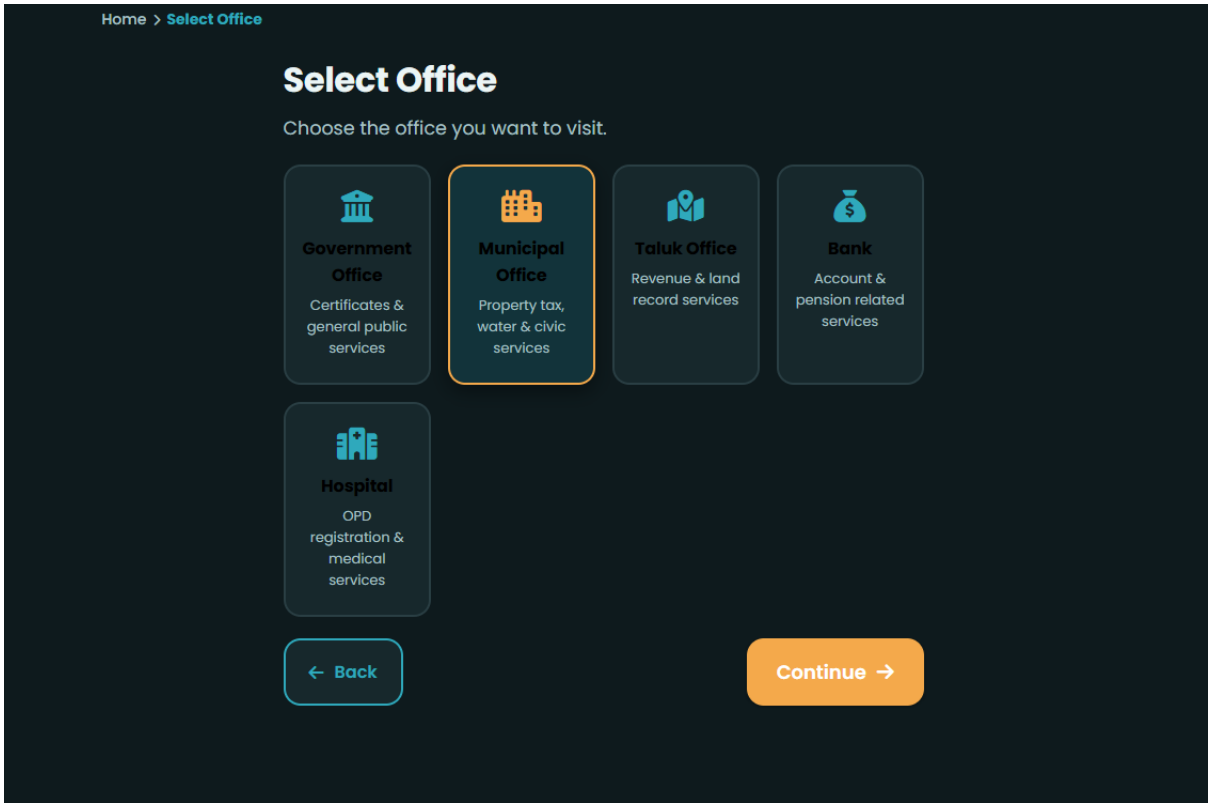
Problem	Cause	User Impact	Fix
4. Counter Number Hidden *(Feedback)*	**Assigned desk only appears on notification modal.**	**Users might know it's their turn but miss the desk number.**	**Pin a banner: "Go to Counter 3" directly on the queue page.**
5. Low Contrast History Chips *(Visibility)*	**Active filter borders blend into dark background.**	**Hard to tell if "Completed" or "All" is selected.**	**Use solid bright pill fills and count badges (e.g., Done: 3).**

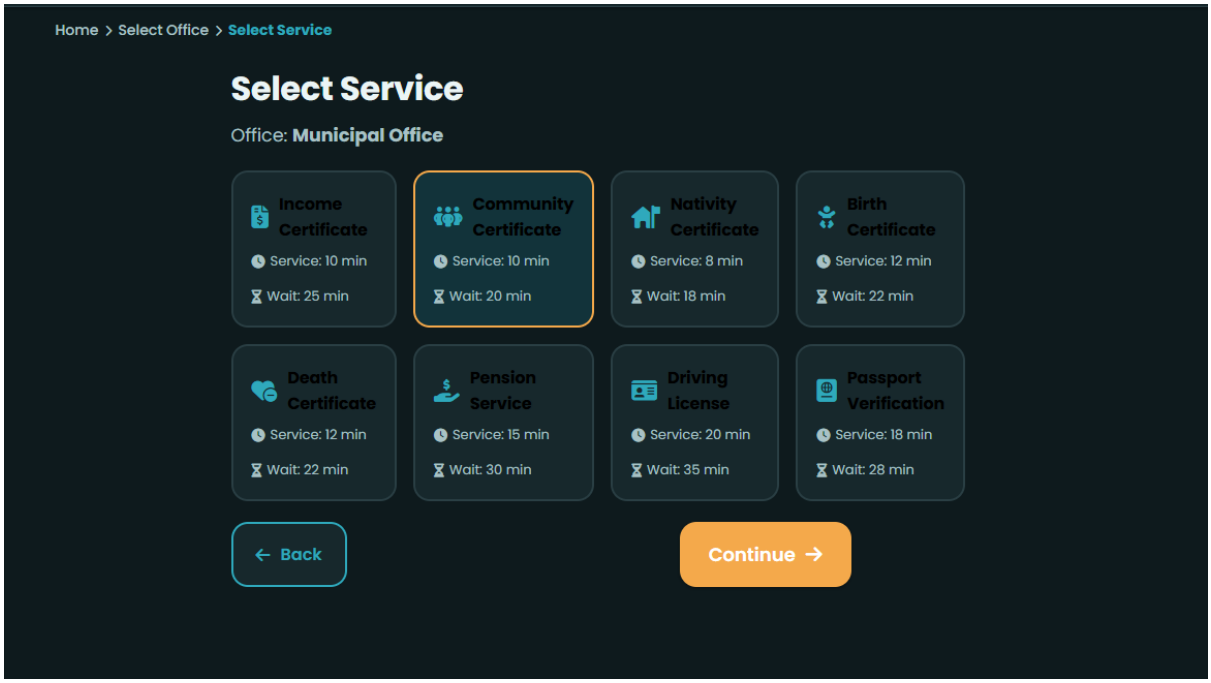
9. Frontend Implementation Evidence

Part E – Frontend Implementation Evidence

- **UI Components Implemented:**
    1. **Interactive Office & Service Selectors (#screen-office, #screen-service):** Dynamic card grids featuring hover elevations, active-state highlighting, and auto-populated time/wait metadata.
    2. **Priority Engine & Token Generator (#screen-priority, #screen-token):** Dynamic SVG score calculations, custom dashed-divider ticket styling, QR code placeholders, and live text-blob download/print triggers.
    3. **Live Queue & Admin Analytics (#screen-queue, #screen-admin-dashboard, #screen-analytics):** Real-time JavaScript simulation intervals, counter management toggles, and custom HTML5 Canvas-rendered charts (Pie, Bar, Line).
- **Navigation & Feedback Handling:** Screen-switching Single Page Application (SPA) architecture with focus management for accessibility, custom toast alert messages, and live screen-reader regions.
- **Key Difference Between Wireframe and Frontend Implementation:**
    - **The Difference:** The original wireframe planned static queue tables, whereas the final frontend implementation features an active timer loop (setInterval) driving a dynamic progress bar, decrementing wait times, and triggering automatic turn-approaching toast notifications.
    - **Justification for Change:** Static displays fail to give waiting users confidence that the system is functioning; live automated status changes provide necessary psychological reassurance in public queue environments.


Welcome to SmartQueue
Book your turn. Skip the wait. Priority for those who need it most.
Book Token
Track Queue
Login
Register
Language
English
தமிழ்
Staff / Admin Login


Home > Select Office
Select Office
Choose the office you want to visit.
Government Office
Certificates & general public services
Municipal Office
Property tax, water & civic services
Taluk Office
Revenue & land record services
Bank
Account & pension related services
Hospital
OPD registration & medical services
← Back
Continue →



10. Critical Reflection

Part F – Critical Reflection

- **1. Multilingual Audio Announcements & Voice Alerts (Web Speech API)**
    - **Improvement:** Integrating native browser-based text-to-speech voice announcements in both English and Tamil for real-time queue changes and turn calls (e.g., *"Token T-118, please proceed to Counter 3"*).
    - **Problem Addressed:** Visually impaired citizens and elderly users who cannot continuously look at a mobile screen or understand visual text cues risk missing their counter call.
    - **Expected User Benefit:** Provides eyes-free, audible queue tracking, reducing anxiety and allowing users to relax in waiting halls without staring at their screens.
    - **Priority: High (Rank 1 – Highest Priority)**
- **2. Offline PWA Caching & Automated SMS Fallback Notification**
    - **Improvement:** Adding Progressive Web App (PWA) service workers to cache generated tokens offline and triggering automated SMS updates at critical queue intervals (5, 2, and 0 tokens ahead).
    - **Problem Addressed:** Government offices, taluk centers, and hospital basements often have weak or fluctuating mobile internet connectivity, causing live WebSocket/timer syncs to drop.
    - **Expected User Benefit:** Ensures the user's token is always viewable offline and delivers turn alerts reliably via standard cellular SMS even without active mobile data.
    - **Priority: Medium (Rank 2)**

- **3. Accessible Document Pre-Verification & Fast-Track Digital Upload**
    - **Problem Addressed:** Priority categories currently rely on self-declaration, which creates bottlenecks and friction when counter staff must manually inspect physical disability or senior citizen certificates.
    - **Expected User Benefit:** Allows users (or their helpers) to snap and attach a photo of their ID/certificate during token booking, speeding up desk clearance by up to 50%.
    - **Improvement:** A lightweight, accessible camera/document upload module during priority selection.
    - **Priority: Low (Rank 3)**

Implementation Choice & Justification

Improvement to Implement First: Multilingual Audio Announcements & Voice Alerts (Web Speech API)

Justification: SmartQueue is fundamentally designed around accessibility for elderly citizens and persons with disabilities. A purely visual interface fails individuals with visual impairments, cataracts, low literacy, or cognitive fatigue from continuous screen monitoring. Implementing localized voice synthesis directly eliminates this exclusion barrier using existing web APIs without requiring any external hardware or complex backend infrastructure, delivering the highest immediate human impact on accessibility.